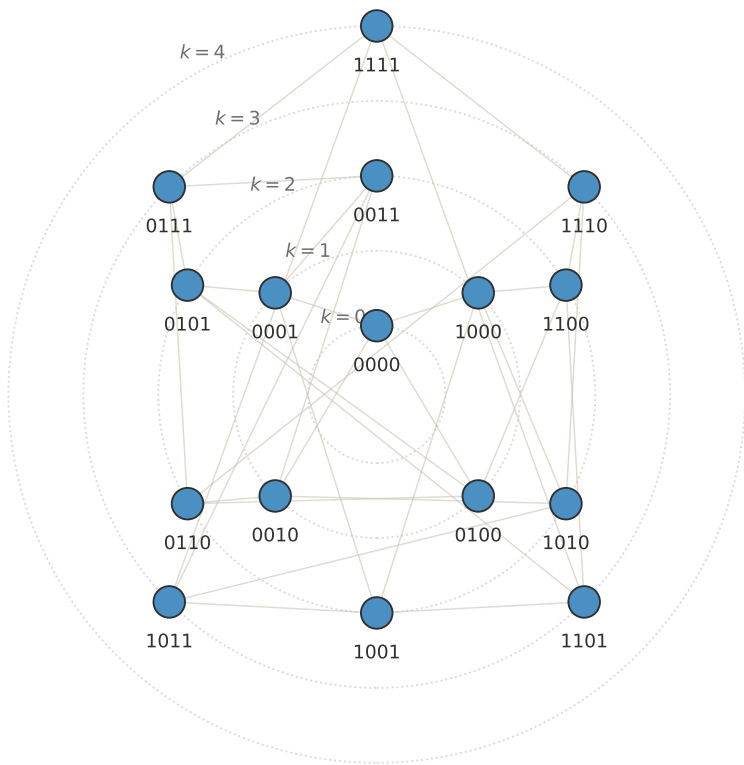


X-mixer vs XY-ring mixer — subspace structure

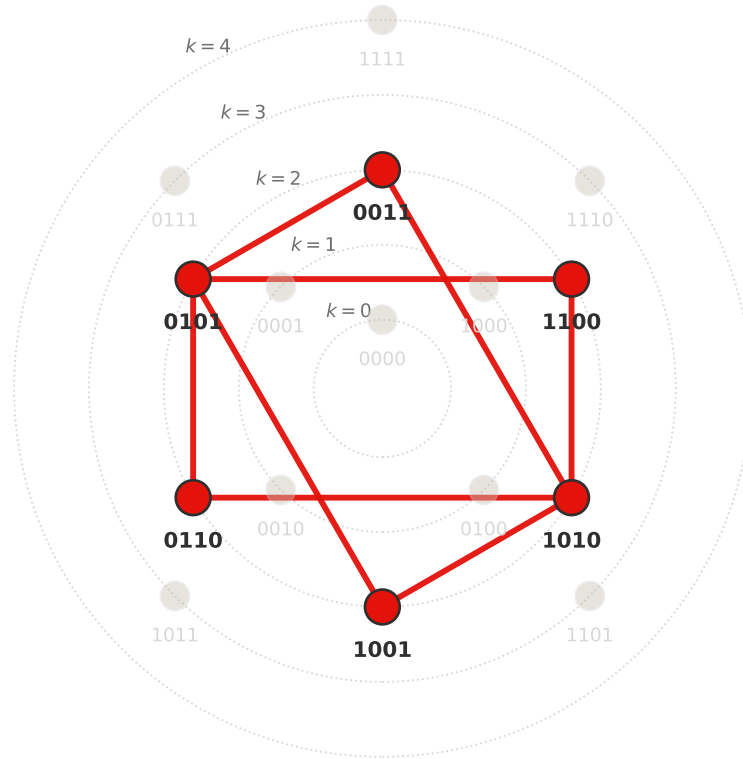
$n = 4$ qubits; rings group bitstrings by Hamming weight $k = \sum_i x_i$

X-mixer — single bit flips



arrows leak across all Hamming weights;
feasibility $\sum_i x_i = K$ is enforced only by the penalty term in H_C

XY-ring mixer — initialised at $|D_K^n\rangle$



dynamics confined to the weight- $k = 2$ ring;
feasibility preserved by construction